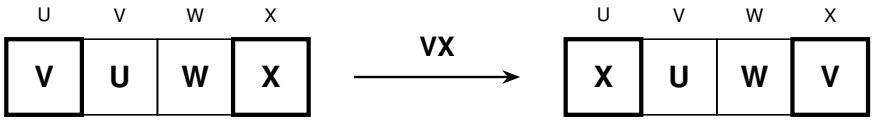
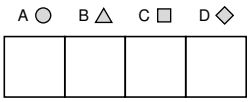
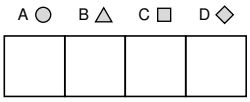
The homes stay in place. A card is at home when it sits in the home with its own letter. A swap trades the places of any two cards, near or far. Write each swap as the letters of the two cards it trades.

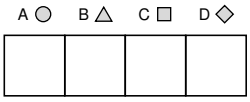


Problem 1: Start with A, B, C and D at home. Swap cards until no card is at home, using as few swaps as you can. Which rows can you end with in that many swaps? Then find the fewest swaps for all five cards.

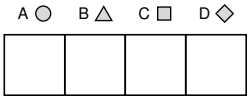
swaps _____ 

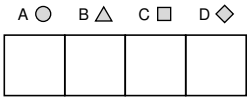
swaps _____ 

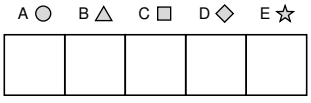
fewest swaps

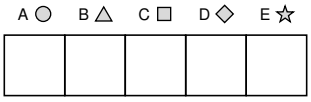
swaps _____ 

four cards ☐

swaps _____ 

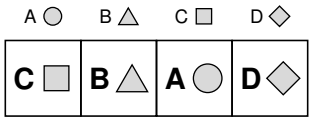
swaps _____ 

swaps _____ 

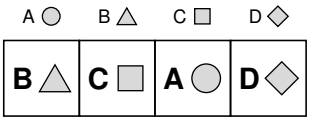
swaps _____ 

five cards ☐

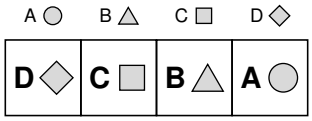
Problem 2: Set out each row below. Bring every card home using as few swaps as you can.



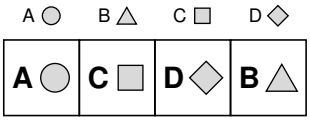
swaps _____ fewest ☐



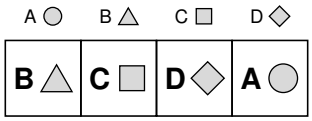
swaps _____ fewest ☐



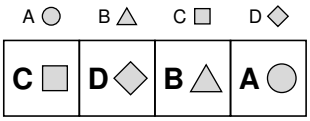
swaps _____ fewest ☐



swaps _____ fewest ☐

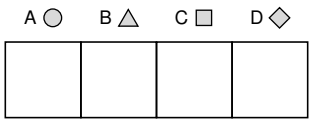


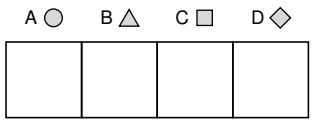
swaps _____ fewest ☐

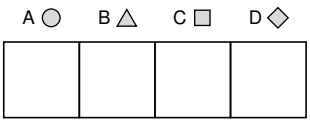


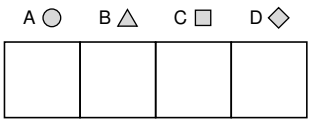
swaps _____ fewest ☐

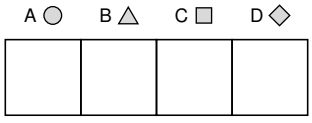
Problem 3: Find every row of A, B, C and D that needs the most swaps to come home. Could any row need 4 swaps? Explain.

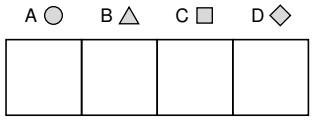


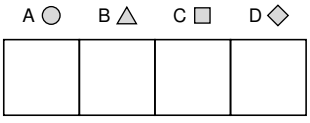


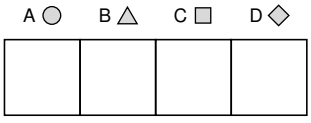




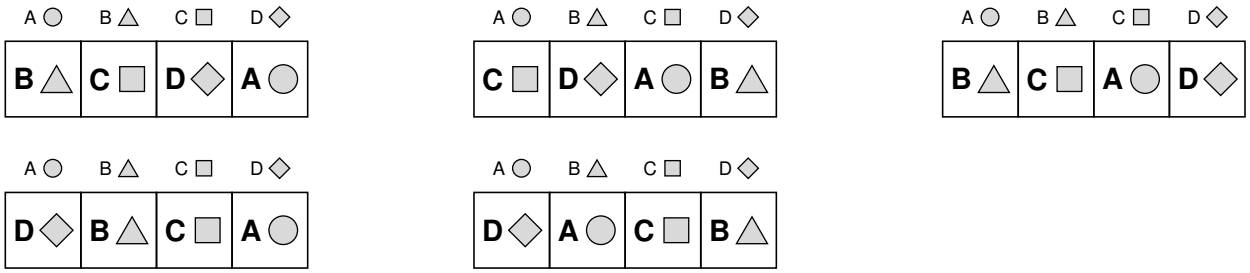








Problem 4: In which of these rows can one swap bring two cards home at once? Explain how you can tell for any row.



Problem 5: On the uncut sheet of 24 A–D outcome cards, circle every letter that sits in its own home. How many circles are there in all? How many does each letter get?

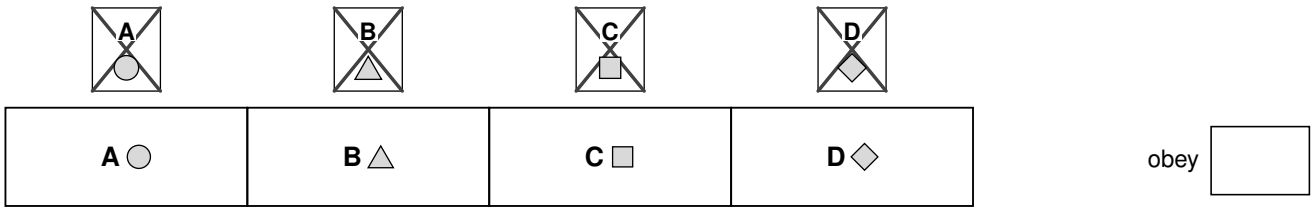
Problem 6: How many circles would the six A–C outcome cards get? There are 120 rows of the five cards A, B, C, D and E. How many circles would all 120 rows get? Explain without listing the rows.

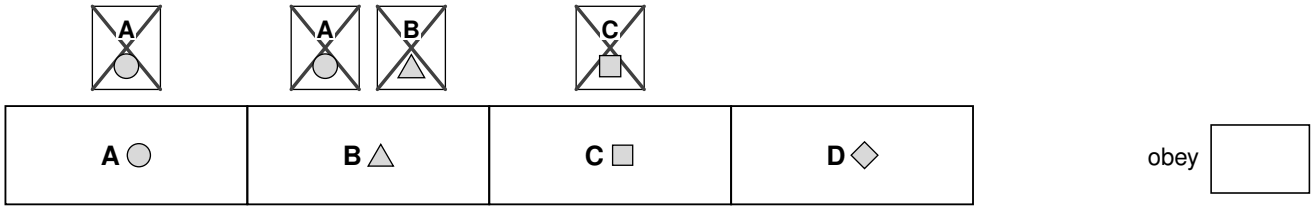
A–C circles ☐ A–E circles ☐

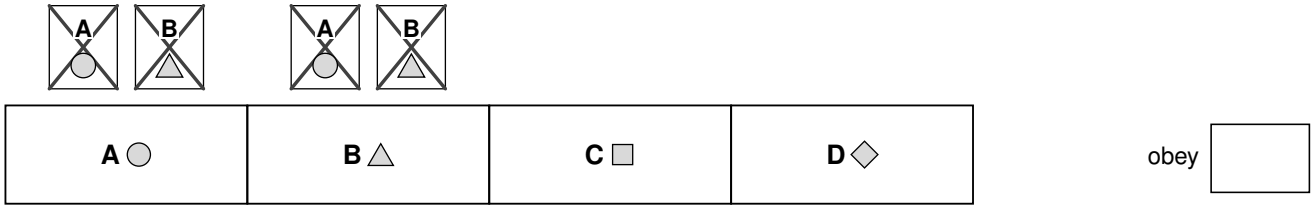
Problem 7: Sort the 24 cut A–D outcome cards into piles by how many letters are at home. Explain why the pile with exactly one letter at home and the pile with exactly two have the sizes they do.

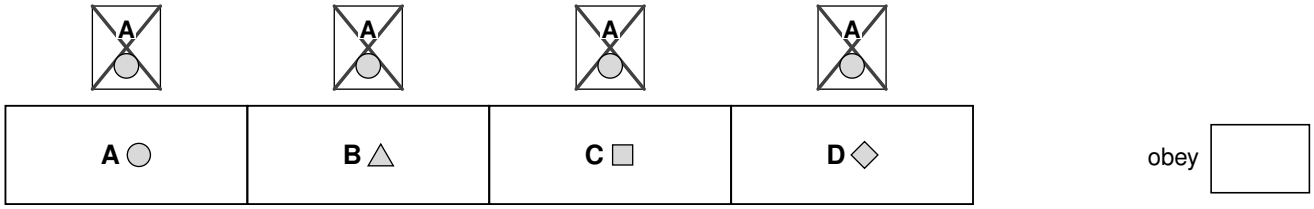
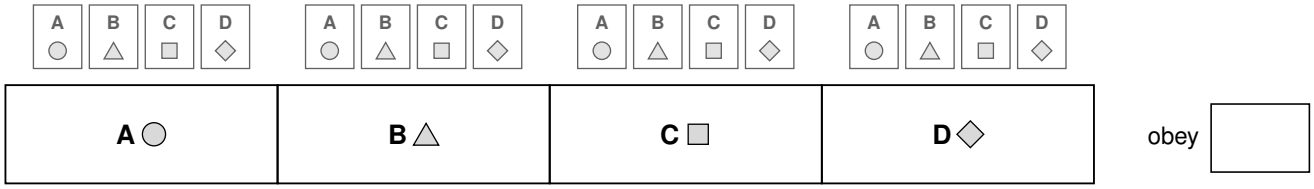
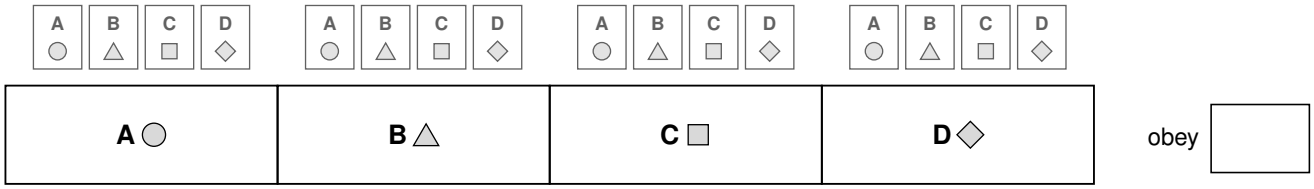
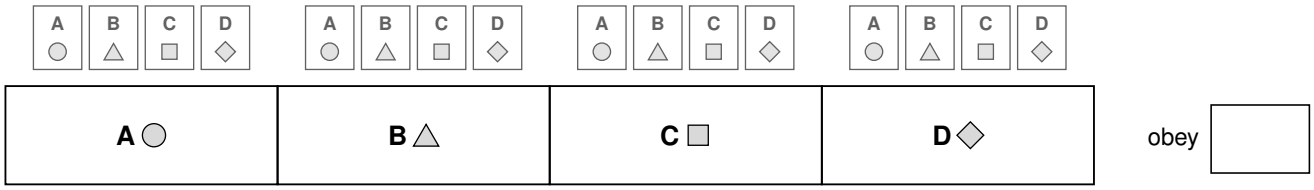
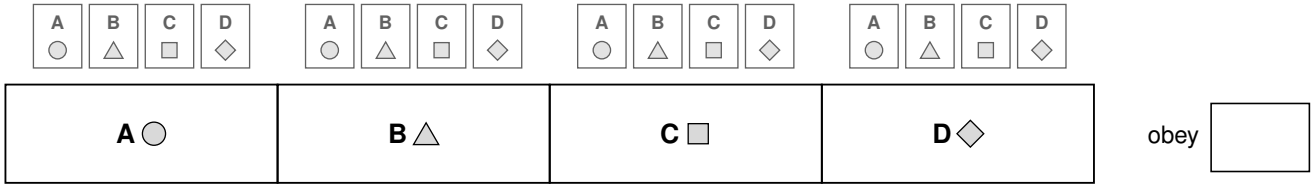
at home	**0**	**1**	**2**	**3**	**4**
how many					

Problem 8: A crossed-out card above a home may not sit in that home. How many of the 24 A–D outcome cards obey each board?









Problem 9: Cross out four cards on the first board so that as many of the 24 outcome cards as possible obey it. On the other boards, cross out four cards so that exactly 8 obey, and then four so that exactly 6 obey. Can four crossed-out cards make exactly 5 obey?